Candidate Seat no………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech (IT) Examination

March-Apr2018

IT217 Digital Logic & Design

Date: 30.03.2018,Friday **Time: 01:30p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1	Answer the question below.	**[07]**
a.	Define a Fan out.	**01**
b.	What is the advantage of ECL family?	**01**
c.	A bubbled NAND gate is equivalent to a _______________ gate.	**01**
d.	$(10101010101)_2 = (\quad)_{16}$	**02**
e.	The voltages in digital electronics are continuously variable. a) True b) False	**01**
f.	Which gates are known as universal gates?	**01**
Q – 2.a	Subtract the following number using 2's complement. 1) $(10)_{10} - (05)_{10}$ 2) $(1010101)_2 - (10101)_2$	**[04]**
Q – 2.b	**Answer any two questions.** 1) State and prove the De Morgan's law using truth table. 2) Design a Full adder combinational circuit. 3) Design a combinational circuit with three inputs and three outputs. The output generates 2's complement of the input binary numbers.	**[10]**
Q - 3	Answer the following.	
a.	Design BCD to Excess-3 code converter.	**[06]**
b.	1) Explain the different type of ROM in detail	**[04]**
	2) Design 4 –input priority encoder.	**[04]**
	OR	
	1) Implement F=(a,b,c) = $\sum m(0,1,3,5,7)$ with the help of multiplexer	**[04]**
	2) Simpify the following Boolean function F , together with the don't care	**[04]**

conditions d, and then express the simplified function in sum of minterms form.

$F(A, B, C, D) = \sum (0,6,8,13,14)$

$d(A, B, C, D) = \sum (2,4,10)$

SECTION – II

Q - 4 Answer the question below. **[07]**

a. The storage element for a static RAM is the ________. **01**

a) Diode b) Register c) Capacitor d) Flip-flop.

b. Give one application of S-R Flip flop. **01**

c. Give the difference between combinational circuit and sequential circuit. **02**

d. What do you mean by don't care condition? **01**

e. Give the difference between ring counter and Johnson ring counter. **02**

Q – 5.a Design 4: 16 Decoder using 2: 4 Decoders. **[05]**

OR

Q – 5.a Design a 2 bit magnitude comparator circuit. **[05]**

Q – 5.b Reduce the number of state in the following state table and draw the reduced state diagram. **[04]**

Present state	Next state		Output	
	X = 0	X = 1	X = 0	X = 1
a	b	c	1	0
b	f	d	0	0
c	d	e	1	1
d	f	e	0	1
e	a	d	0	0
f	b	c	1	0

Table:1

Q – 5.c Draw the Johnson counter which can generate 6 different timing signals. Also mention the timing sequence with respective required decoding. **[05]**

OR

Q – 5.c Derive a state table , state diagram and state equation for the given sequential circuit in Figure 1. **[05]**

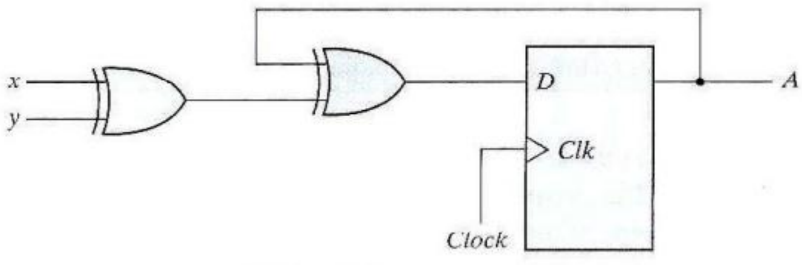

Figure 1.

Q – 6.a Make a characteristic table and excitation table for RS, JK and T flip-flops. **[03]**

Q – 6.b Design a 3 bit binary counter which will count in descending order. **[07]**

Q – 6.c Give the importance of master slave Flip flop and draw logic diagram of master slave J-K flip flop. **[04]**

OR

Q – 6.a Design a counter with the following binary sequence: 0, 1, 3,7,6,4 and repeat. Use T flip flop for designing. **[07]**

Q – 6.b Draw and explain the 2 bit Universal shift counter. **[04]**

Q – 6.c Give the difference between synchronous and asynchronous counter. **[03]**

***